WORKED SOLUTION

STEP II Specimen paper, Question 2

Pure • Symmetric Simultaneous Equations

STEP II Specimen paper, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q2

The key idea

The three equations are cyclic, so subtracting them in pairs is natural, and each difference factorises with $x+y+z$ as a common factor. The sum of squared differences is then useful twice over: in terms of x, y, z it is a multiple of $a+b+c$, while the first result expresses it in terms of $a-b$, $b-c$, $c-a$ and $x+y+z$. Once we know $x+y+z$ and the pairwise differences, three linear equations finish the job.

Solution

The equal ratios

Subtract the second equation from the first:

$$\begin{aligned} a-b &= x^2-y^2-yz+zx \\ &= (x-y)(x+y)+z(x-y) \\ &= (x-y)(x+y+z). \end{aligned}$$

In exactly the same way (or by cycling $x \to y \to z \to x$, $a \to b \to c \to a$),

$$b-c=(y-z)(x+y+z), \qquad c-a=(z-x)(x+y+z).$$

Write $S=x+y+z$. Since $a \neq b$, the product $(x-y)S$ is non-zero, so $S \neq 0$ (and likewise x, y, z are distinct). As a, b, c are distinct we may divide by $a-b$, $b-c$, $c-a$ and by S, giving

$$\frac{y-z}{b-c}=\frac{z-x}{c-a}=\frac{x-y}{a-b}=\frac{1}{x+y+z},$$

as required.

Finding $x+y+z$

Expand the suggested sum:

$$\begin{aligned}(y-z)^2+(z-x)^2+(x-y)^2 &= 2(x^2+y^2+z^2-xy-yz-zx)\\ &= 2[(x^2-yz)+(y^2-zx)+(z^2-xy)]\\ &= 2(a+b+c).\end{aligned}$$

On the other hand, the first part gives $y-z=(b-c)/S$ and so on, so the same sum equals

$$\frac{(b-c)^2+(c-a)^2+(a-b)^2}{S^2}=\frac{2(a^2+b^2+c^2-bc-ca-ab)}{S^2}.$$

Equating the two expressions and dividing by $2(a+b+c)$, which is positive because a, b, c are positive,

$$S^2=\frac{a^2+b^2+c^2-bc-ca-ab}{a+b+c}.$$

So $x+y+z=\triangle$ where $\triangle^2$ is this quantity, as required. Note that $\triangle^2>0$, since its numerator is half of $(b-c)^2+(c-a)^2+(a-b)^2$, which is positive for distinct a, b, c. So there are two possibilities, $\triangle=\pm\sqrt{\triangle^2}$, and nothing so far picks one of them.

Solving the equations

We now know

$$x+y+z=\triangle,\qquad x-y=\frac{a-b}{\triangle},\qquad x-z=\frac{a-c}{\triangle}.$$

Adding all three,

$$3x=\triangle+\frac{2a-b-c}{\triangle}=\frac{\triangle^2+2a-b-c}{\triangle}.$$

This simplifies nicely. Put $s=a+b+c$ and $Q=a^2+b^2+c^2-bc-ca-ab$, so $\triangle^2=Q/s$. Then

$$\begin{aligned}s(\triangle^2+2a-b-c) &= Q+(a+b+c)(2a-b-c)\\ &= (a^2+b^2+c^2-bc-ca-ab)+(2a^2+ab+ca-b^2-c^2-2bc)\\ &= 3(a^2-bc).\end{aligned}$$

Hence $x=\dfrac{a^2-bc}{s\triangle}$, and y, z follow by cycling. Now $(s\triangle)^2=sQ=a^3+b^3+c^3-3abc$, by the standard factorisation

$$a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-bc-ca-ab).$$

Writing $K=\sqrt{a^3+b^3+c^3-3abc}>0$, we have $s\triangle=\pm K$, and the solutions are

$$(x,y,z)=\pm\frac{1}{K}(a^2-bc,\ b^2-ca,\ c^2-ab),$$

with the same sign throughout.

Ask yourself We only showed that any solution must be of this form. Are both of these really solutions?

Yes. Everything above was a chain of consequences, so we should substitute back. Since only squares and products of two of the unknowns appear, the overall sign doesn't matter, and

$$\begin{aligned}(a^2-bc)^2-(b^2-ca)(c^2-ab) &= a^4-2a^2bc+b^2c^2-(b^2c^2-ab^3-ac^3+a^2bc)\\ &= a(a^3+b^3+c^3-3abc) = aK^2.\end{aligned}$$

Dividing by K^2 gives $x^2-yz=a$; the other two equations follow by cycling. So there are exactly two solutions, one the negative of the other.

Checks

- Take $a=1$, $b=2$, $c=3$. Then $a^3+b^3+c^3-3abc=18$, $K=3\sqrt{2}$, and the solution with the plus sign is $(x,y,z)=\frac{1}{3\sqrt{2}}(-5,1,7)$. For example $x^2-yz=\frac{25-7}{18}=1$, and $x+y+z=\frac{1}{\sqrt{2}}$, whose square is $\frac{1}{2}=\frac{14-11}{6}=\triangle^2$.
- A computer tested 20,000 random triples of distinct positive a, b, c, with both signs: the formulas satisfied all three equations and the equal ratios, with relative error below 10^{-11}.

Common slips

- Taking only the positive square root for $\triangle$: the question's $\triangle$ is defined only through $\triangle^2$, and both signs give genuine solutions.
- Dividing by $x+y+z$ without saying why it is non-zero. Here it follows from $a\neq b$.

Beyond STEP

The identity $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ is the determinant of the circulant matrix with rows (a,b,c), (c,a,b), (b,c,a), and it factorises completely over the complex numbers as $(a+b+c)(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c)$ with $\omega=e^{2\pi i/3}$. In fact the map $(x,y,z)\mapsto(x^2-yz,\,y^2-zx,\,z^2-xy)$ is the adjugate of a circulant matrix, which explains why it is so easily inverted. Circulant matrices and their eigenvectors are a gentle first meeting with the discrete Fourier transform; Strang's *Introduction to Linear Algebra* is a good place to read more.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.